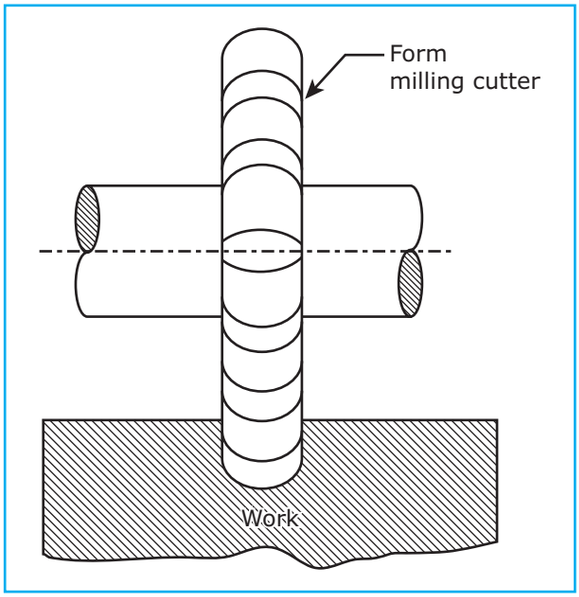
Form Milling

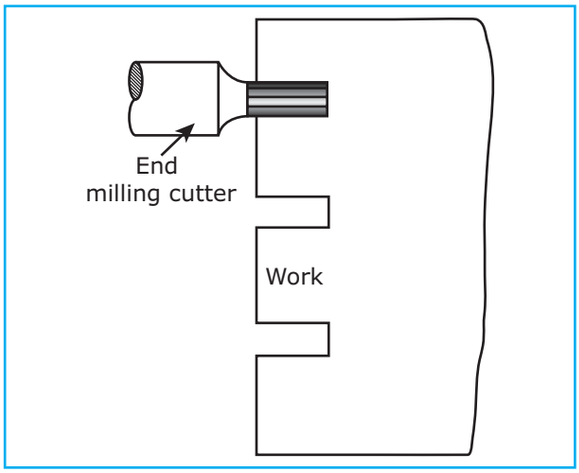
The form milling is the operation of production of irregular contours by using from cutters. Machining convex and concave surfaces and gear cutting are some examples of form milling.



Form milling operation.

End milling

It is the operation of producing a flat surface which may be vertical, horizontal or at an angle to the table surface. The end milling is performed by a cutter known as an end mill. End milling is mostly performed in a vertical machine.



End milling operation

Keyway milling

The operation of production of keyways, grooves and slots of different shapes and sizes can be performed in a milling machine by using a plain milling cutter, a metal slitting saw, an end mill or by a side milling cutter.

Gear cutting

Gear cutting operation is performed in a milling machine by using a form cutter. The work is held between centers on a universal dividing head. A proper gear cutter is selected and the teeth are cut by DP, module method.

Cam milling

Cam milling is the operation of producing cams in a milling machine with the use of a universal dividing head and a vertical milling attachment. It is performed by end mills on the cam blank.

5.16 Cutting speed, feed and depth of cut

Cutting speed

It is the distance travelled by a point on the cutting edge of the milling cutters to remove metal in time duration of one minute. It is expressed in meters per minute.

$$\text{Cutting speed} = \frac{\pi dn}{1000} \text{ meter/minute}$$

$$\pi = \frac{22}{7} \text{ (or) } 3.14,$$

Where, D = The diameter of the milling cutter in mm

N = Spindle speed in rpm.

The cutting speed depends upon the material to be machined, the cutter

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